<u>EXPERIMENT 10</u>

Iris Flower Classification using K-Nearest Neighbors (KNN)

1. Problem Definition

The objective of this experiment is to classify iris flowers into three species:

- **Setosa (Class 0)**
- **Versicolor (Class 1)**
- **Virginica (Class 2)**

This is a **multiclass classification problem**.

The goal is to build a model that can accurately predict the species based on flower measurements.

2. Data Understanding

The dataset used is the **Iris dataset** from sklearn.

Dataset Characteristics

- Total samples: **150**
- Features: **4**
  - Sepal Length
  - Sepal Width
  - Petal Length
  - Petal Width
- Classes: **3 (balanced)**
  - Each class has **50 samples**

Important Insight:

- Dataset is **perfectly balanced**
- No class imbalance issues

<u>Exploratory Data Analysis (EDA)</u>

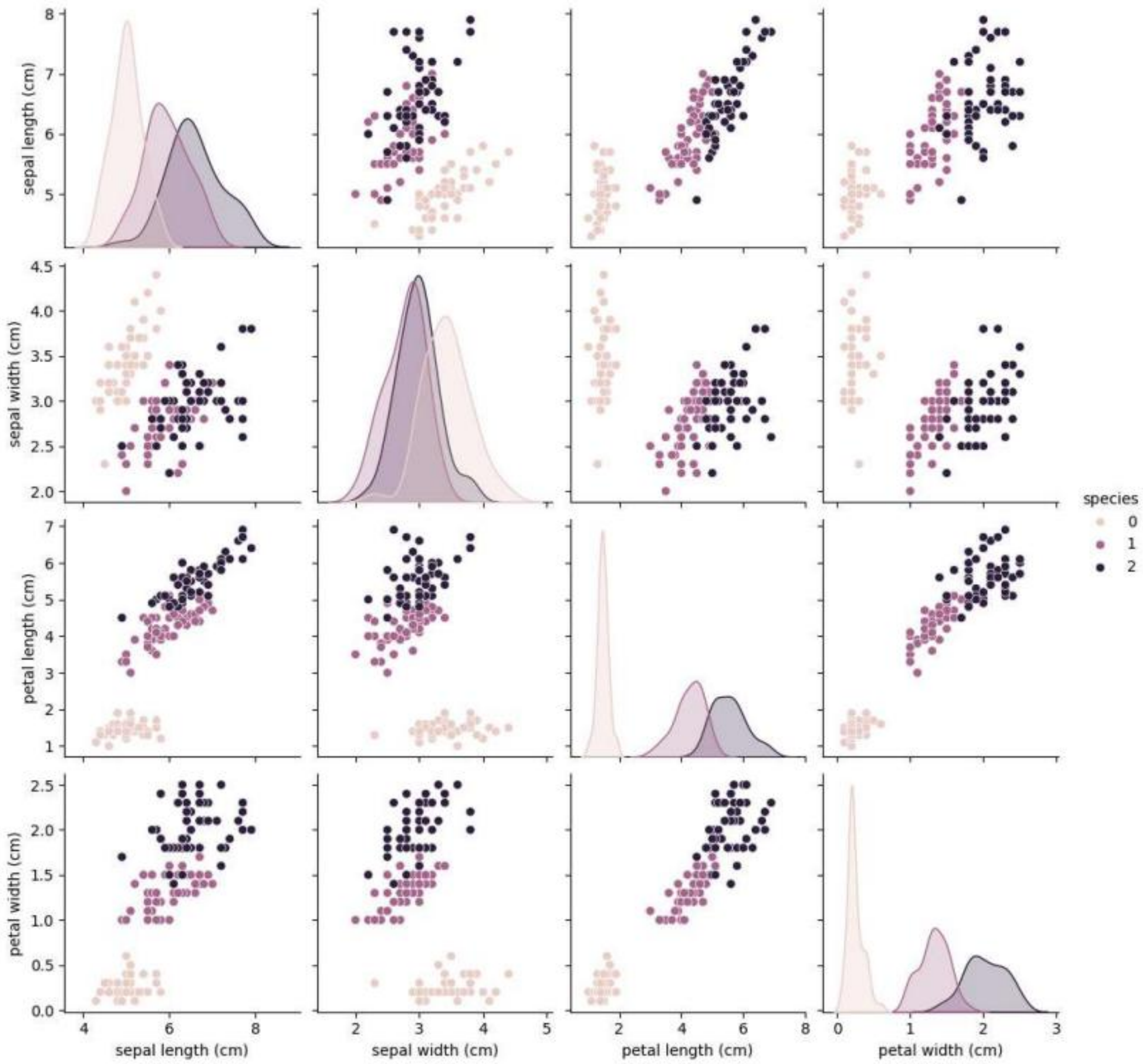
From pairplot visualization:

Observations:

- **Setosa is completely separable**
- Versicolor and Virginica overlap slightly
- Petal features provide strongest separation

Interpretation:

- Dataset is **linearly separable (almost)**
- Easy classification problem



3. Train–Test Split

- 80% Training Data
- 20% Testing Data

Ensures evaluation on unseen samples.

4. Model Used: K-Nearest Neighbors (KNN)

Working Principle:

- Finds **k nearest data points**
- Assigns class based on majority vote

Parameter:

- **k = 3**
- Small k : sensitive to noise
- Large k : smoother decision boundary

5. Evaluation and Prediction

Accuracy

$$\text{Accuracy} = 1.0$$

Model correctly classified **all test samples**

Confusion Matrix

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

Classification Report

- Precision = 1.00
- Recall = 1.00
- F1-score = 1.00

for all classes.

<u>Prediction on New Data</u>

Input sample:

[5.1, 3.5, 1.4, 0.2]

Prediction:
Setosa

Correct because:

- These feature values match Setosa cluster

6. Model Analysis

1. Dataset is Simple

- Clear separation between classes
- Especially Setosa

2. Low Dimensional Data

- Only 4 features
- Easy decision boundaries

3. Balanced Dataset

- No bias toward any class

4. KNN Works Well for Small Data

- Stores all data
- Uses distance → effective here

Best Model Justification

KNN is effective here because:

- Data is clustered clearly
- Distance-based classification works well